

Assessing significant geomorphic changes and effectiveness of dynamic restoration in a coastal dune ecosystem

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ABSTRACT

A shift from restoring coastal dunes as stabilized landscapes toward more morphodynamic ecosystems is underway. This paper uses results from a recent case study where invasive vegetation was removed from a coastal dune complex in western Canada as a first step in a dynamic ecosystem restoration project. Spatial statistical methods, used in the natural sciences to quantify patterns of significant spatial–temporal changes, are reviewed and the local Moran's I_i spatial autocorrelation statistic is explored for detecting and assessing significant changes. Cluster maps of positive (depositional) and negative (erosional) changes were used to derive statistically significant volumetric changes within discrete geomorphic units (beach, foredune, transgressive dune) over one year following vegetation removal. All units experienced net increases in sediment budgets compared to a pre-restoration surface. The beach experienced the highest episodic erosion and volumetric change and greatest net annual sediment budget. Compared to the beach, the annual sediment budget of the foredune was 19% whereas the transgressive dune was 33%. The foredune recovered rapidly to initial erosion during restoration and subsequent natural events with consistently positive sediment volumes and attained a form similar to that pre-restoration. Aeolian deflation and sand bypassing through the foredune was greatest in the two months following vegetation removal and peak accretion in the transgressive dune resulted from depositional lobes extending from the foredune, smaller dunes migrating within the complex, and growth of a precipitation ridge along the eastern margin.

Several methodological and logistical considerations for detecting significant change in dynamic dune landscapes are discussed including sampling strategy design, data normalization and control measures, and incorporating uncertainty and inherent spatial relations within acquired datasets to ensure accuracy and comparability of results. Generally underutilized in coastal geomorphology, spatial autocorrelation methods (e.g., local Moran's I_i) are recommended over spatially uniform threshold approaches for the ability to detect local change processes and explore hypotheses on spatial–temporal dynamics.

Finally, several key geomorphic indicators, that are believed to aid in re-establishing ecological conditions and processes that favor more resilient and natural dune ecosystems, are identified for assessing the effectiveness of dynamic restoration projects including: increased aeolian activity, enlarged active sand surface area, positive sediment budgets, increased dune morphodynamics, improved geomorphic diversity, and enhanced geomorphic resilience. Although limited in temporal scope, the case study results show that the initial phase of the restoration treatment was effective in enhancing all indicators except for increasing sand surface area. Given decadal scale observations of climatic changes and longer-term eco-geomorphic trajectory toward stabilization in the region, however, it is unlikely that the geomorphic effectiveness of this restoration effort will continue without continued frequent treatment interventions.

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1. Introduction

In areas with appreciable onshore sand supply, foredunes are a significant component of the coastal sediment budget as they store and cycle substantial amounts of sand in the backshore (e.g., Short and Hesp, 1982; Psuty, 1988; Hesp, 2002; Psuty, 2004). As such, coastal dunes provide an important buffer that can protect shorelines against storm surge

flooding, coastal erosion, and more gradual sea-level rise (e.g., Davidson-Arnott, 2005; Houser et al., 2008; Mascarenhas and Jayakumar, 2008; Eamer and Walker, 2010). Coastal dunes are also ecologically significant as they provide critical habitat for many specialized endemic, migratory and endangered species (e.g., Wiedemann and Pickhart, 1996; Wiedemann, 1998; Grootjans et al., 2002; Hesp, 2002) and serve as an important natural resource and land use base for coastal development (e.g., Nordstrom, 1990; Riksen et al., 2006; Nordstrom, 2008).

Traditionally, coastal dune systems have been restored to a 'stabilized' state so as to halt natural geomorphic processes of erosion, sand

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drift, and dune migration. This has often involved planting non-native plants, shrubs, or trees or by physically hardening or armouring dune features. Such stabilization efforts have been implemented for a variety of purposes including forestry and agriculture (e.g., Van Der Meulen and Salman, 1996; Riksen et al., 2006), urban and recreational development (e.g., Riksen et al., 2006), groundwater storage and recharge (e.g., Arens et al., 2004; Arens and Geelen, 2006), and flood protection and wave erosion defense (e.g., Hillen and Roelse, 1995; Arens et al., 2001; Grootjans et al., 2002; Arens et al., 2004; Mascarenhas and Jayakumar, 2008). Over the last three decades, however, coastal researchers and managers have recognized that efforts of dune stabilization have resulted in the loss of landform dynamics, complexity, and resilience. In turn, this has entailed notable ecological impacts, such as declines in early successional floral species and a corresponding loss of species richness and diversity (e.g., Grootjans et al., 2002; Arens et al., 2004; Hilton et al., 2005; Nordstrom, 2008). Artificially stabilized dune systems are often resistant to all but the most extreme disturbances and, as a result, have dysfunctional geomorphic and ecological regimes that do not experience lower magnitude disturbance cycles required for maintaining natural dune ecosystem structure and function (Nordstrom, 1990, 2008).

Re-establishment of natural disturbances and related morphodynamics in dune landscapes are being incorporated increasingly into restoration projects that seek to restore lost ecosystem dynamics and services. Recent work suggests that a more dynamic landscape, wherein natural geomorphic processes are stimulated, provides a more resilient ecosystem with more favorable ecological conditions for native communities and endangered species. For instance, restoration projects are seeking to reactivate aeolian activity and dune mobility via vegetation removal by fire, herbicides, mechanical or manual pulling, or soil tillage that encourage a more dynamic landscape and associated ecosystems (e.g., van Boxel et al., 1997; Nordstrom et al., 2002; Arens et al., 2004; Rozé and Lemauiel, 2004; Van Der Meulen et al., 2004; Arens and Geelen, 2006; Nordstrom, 2008; Hilton et al., 2009; Kollmann et al., 2011). This 'dynamic' approach to restoration effectively enhances aeolian activity and dune morphodynamics to produce a more diverse landscape with more periodic erosion–stabilization cycles that, in turn, provides required environmental conditions and gradients required by natural ecological communities (Kooijman, 2004; Van Der Meulen et al., 2004; Arens and Geelen, 2006; Nordstrom, 2008). Although dynamic restoration approaches are not yet conventional, a paradigm shift from decades of dune landscape stabilization and ensuing ecological dysfunction toward more dynamic, disturbance resilient approaches appears to be underway.

Dynamic restoration projects provide distinct research opportunities to quantify and interpret resulting sediment transfers and morphodynamic responses in coastal dune ecosystems. In turn, this can provide useful insights into the effectiveness and refinement of implemented disturbances and treatment regimes. To date, much research on the restoration of dynamic coastal dunes is predominantly ecological and focuses on soil and vegetation changes (e.g., van Boxel et al., 1997; Ketner-Oostra and Sykora, 2000; Grootjans et al., 2002). Geomorphic research has concentrated primarily on measuring changes in active sand surface area or cross-shore topographic profiles with only indirect measures of aeolian activity (e.g., Nordstrom et al., 2002; Arens et al., 2004, 2005; Wondergem, 2005; Arens and Geelen, 2006; Hilton et al., 2009). This research has provided foundational knowledge of the responses of dune landscapes to dynamic restoration. Recent investigations in other areas of physical geography (e.g., Luoto and Hjort, 2006; Thompson et al., 2006; Adelsberger and Smith, 2009; Wheaton et al., 2010) and spatial ecology (e.g., Wulder et al., 2007; Nelson and Boots, 2008), however, have applied more robust spatial analysis methods to quantify, detect, and interpret significant patterns of change in landscapes that have direct relevance

and utility for assessing the effectiveness of dynamic restoration treatments in coastal dune ecosystems.

In response to this opportunity, this paper reviews established spatial statistical methods that can be used to quantify and examine significant spatial–temporal volumetric and geomorphic changes within dynamic coastal dune landscapes. The utility of one particular method, local Moran's I_i , for detecting and assessing the impacts of dynamic restoration is demonstrated by a case study where invasive vegetation was removed from a foredune-transgressive dune complex. From this, various methodological and logistical considerations for detecting significant changes in dynamic dune landscapes are discussed and several key geomorphic indicators that can be used to assess the effectiveness of dynamic restoration methods are presented.

2. Quantifying and detecting significant geomorphic changes within dune landscapes

Traditional methods for examining geomorphic changes and related sediment transfers within beach-dune systems have involved interpretation of historical aerial photography (e.g., Tsoar and Blumberg, 2002; Mathew et al., 2010; Heathfield and Walker, 2011), analysis of cross-shore topographic profiles (e.g., Morton et al., 1994; Davidson-Arnott and Law, 1996; Aagaard et al., 2004), monitoring of erosion/deposition pins or quadrat plots (e.g., Gares, 1992; Davidson-Arnott and Law, 1996; Arens et al., 2004; Levin et al., 2006; Ollerhead et al., 2013), and interpretation of digital elevation models (DEMs) derived from detailed repeat topographic surveys (e.g., Gares et al., 1996; Arens, 1997; Andrews et al., 2002; Ruz and Meur-Ferec, 2004; Anthony et al., 2006, 2007) or, more recently, high resolution aerial LiDAR data (e.g., Woolard and Colby, 2002; Sallenger et al., 2003; Houser and Hamilton, 2009; Saye et al., 2005; Eamer and Walker, 2010; Houser and Mathew, 2011). Data from these methods are commonly used to generate estimates of active sand surface area and/or volumetric changes of sediment and related geomorphic responses (e.g., beach-dune erosion and rebuilding, incipient dune growth, dune migration rates). Such results are very useful for interpreting landform to landscape scale responses of beach-dune systems to natural disturbances such as wave erosion by storms, hurricanes, or climatic variability events (e.g., Allan et al., 2003; Ruz and Meur-Ferec, 2004; Anthony et al., 2006; Houser et al., 2008; Houser, 2009; Houser and Hamilton, 2009) as well as to implemented disturbances for restoration purposes (e.g., Nordstrom et al., 2002; Arens et al., 2004, 2005; Wondergem, 2005; Arens and Geelen, 2006; Hilton et al., 2009).

Robust and repeatable methods that account for uncertainty are required to distinguish between noise in acquired DEM datasets and changes that are statistically significant. DEM precision and accuracy are a function of a variety of fundamental factors, including the quality of survey point data (a function of instrument precision), sampling strategy and point density, sampling frequency and temporal consistency, surface composition (e.g., soft sand vs. stable soils), topographic complexity, and chosen interpolation methods (e.g., Wise, 1998; Wechsler, 2003; Wechsler and Kroll, 2006; Heritage et al., 2009; Wheaton et al., 2010). Furthermore, when calculating change surfaces from DEMs, error resulting from uncertainty is additive as a result of comparison of DEMs with individual uncertainties. No standard convention exists for considering and incorporating uncertainty, as evident in recent research on beach-dune morphological changes derived from DEMs (e.g., Woolard and Colby, 2002; Mitsova et al., 2005; Anthony et al., 2006; Mathew et al., 2010), which implements different methods of data acquisition (e.g., LiDAR, RTK-GPS, digital photogrammetry, laser total station surveys) and spatial interpolation models (e.g., inverse distance weighted, regularized spline with tension, kriging) each with respective uncertainties and handling of error. In some cases, data uncertainties are not specified

(e.g., Mitasova et al., 2005) or are assigned a minimum detection threshold (often 5 cm vertical accuracy, e.g., Anthony et al., 2006). If uncertainties are unspecified or unexplored, reported volumetric changes and patterns might include a component that is statistically insignificant and misleading, whereas assigned thresholds may filter out small, but real, changes and patterns. Thus, careful consideration of uncertainty and inherent spatial relations within topographic datasets is required to detect significant changes beyond those of natural variability. This is necessary to ensure the accuracy and comparability of volumetric and geomorphic change results that relate to real process–response dynamics in the landscape.

2.1. Application of spatial statistics for landscape change detection

Spatial statistics are formal analytical techniques developed to measure and model spatial–temporal patterns, processes, and relations in geographic datasets based on the principle of spatial autocorrelation (SA). Spatial autocorrelation describes the amount of dependency or co-variance of an attribute within a geographic area, or spatial neighborhood. Geostatistics (a subset of spatial statistics) are methods used to spatially interpolate, visualize, and model geographic data from point observations. In geomorphology, geostatistical interpolation methods are often used to model the spatial structure and trends in spatially discontinuous data (e.g., x, y, z-elevation measurements) to produce a continuous, representative surface (DEM) at a given point in time. Consideration of data sampling strategy, geostatistical properties within datasets, and uncertainty is critical for the development of accurate DEMs (e.g., Chappell et al., 2003a; Heritage et al., 2009; Wheaton et al., 2010). Extensive reviews of methodological and analytical considerations for DEM modeling in geomorphology are provided elsewhere (e.g., Heritage et al., 2009; Wheaton et al., 2010; Evans, 2011; Milan et al., 2011; Bishop et al., 2012; Wilson, 2012) and were explored in the 41st Binghampton Symposium in Geomorphology (see James et al., 2012).

To assess spatial–temporal dynamics in landscapes (e.g., change detection maps and/or volumetric estimates), accurate spatial models (interpolated DEMs) must be compared over time. The simplest representation of change between two DEMs is to subtract pixel or grid values to create a difference surface. This assumes, however, that modeled DEMs are accurate and that some threshold to define uncertainty, or error, is incorporated. Traditionally, arbitrary thresholds (e.g., based on instrument precision) have been used to differentiate substantive change from background noise, as discussed above. More advanced geostatistics can be used, however, to better characterize spatial associations and uncertainty inherent in geomorphic datasets and, by way of deterministic interpolation models (e.g., weighted least squares, inverse distance weighted, kriging), offer improved predictability and minimize the use of arbitrary parameters. Spatial structure models, or variograms, associated with these methods are useful for quantifying and reducing errors in DEM generation (e.g., Desmet, 1997; Chappell et al., 2003a). Variograms have an observed (experimental) and modeled component. The former describes inherent spatial variation in the dataset and related parameters (e.g., sill, range, nugget, per Swales, 2002) that are used to develop a best approximation model variogram. The model variogram has mathematically uniform properties that enable it to estimate a more continuous surface at unsampled locations to create a DEM. Such methods can be applied to other geomorphic attributes (e.g., sediment transport, erodibility), which is useful for interpreting process–response relations operating at different spatial and temporal scales (e.g., Oliver et al., 1989a; Swales, 2002; Chappell et al., 2003a, 2003b). Application of geostatistics to repeat DEMs for detection and analysis of spatial–temporal patterns in geomorphology has been increasing, particularly at the meso-scale (e.g., Andrews et al., 2002; Swales, 2002; Woolard and Colby, 2002; Chappell et al., 2003a, 2003b; Anthony et al., 2006; Heritage et al.,

2009; Wheaton et al., 2010; Milan et al., 2011). This work explores the utility of examining spatial scales, uncertainty, and inherent spatial associations for change detection in coastal dune systems beyond approaches that use simple, spatially uniform, and often arbitrarily defined thresholds.

2.1.1. Spatial–temporal change detection using Moran's I_i

Spatial pattern analysis of change surfaces provides an approach to quantifying changes that are statistically unexpected based on hypotheses of random change in the landscape (Wulder et al., 2007). Statistical measures of spatial autocorrelation (SA) have gained traction in the natural sciences as a method for quantifying spatial patterns in landscape phenomena (Anselin, 1995; Ord and Getis, 2001; Nelson and Boots, 2008) and assume that spatial patterns emerge as a result of some driving process(es). For instance, local Moran's I_i can be used to map statistically significant SA based on attribute values in relation to neighboring values (Nelson and Boots, 2008). Local measures of SA can be expressed as a cross-product statistic wherein the observed values x_i , $i \in \{1, \dots, n\}$ of a random variable X (e.g., elevation) measured at a set of n data sites, is expressed by the equation

$$I_i = \sum_j w_{ij} y_{ij}, \quad (1)$$

where w_{ij} is a measure of the spatial relationships of data sites i and j at a given time and y_{ij} is a measure of the relationship in attribute space (Getis and Ord, 1992; Boots, 2002). Local Moran's I_i expresses the attribute relationship in the cross-product statistic as

$$y_{ij} = (x_i - \bar{x})(x_j - \bar{x}) \quad (2)$$

and the resulting measure of SA by

$$I_i = \left(\frac{z_i}{\sum_i \frac{z_i^2}{n}} \right) \sum_j w_{ij} z_j, \quad (3)$$

where $z_i = (x_i - \bar{x})$ and w_{ij} defines the spatial relationship between locations i and j . In geomorphic (elevation) datasets, y_{ij} describes the relationship in attribute space between locations i and j , or the change in surface elevation between time periods (or DEMs), and z_i reflects the difference between the change in elevation at a measurement point (local) and the mean (global) change in elevation for the entire study area. As such, Moran's I_i provides a spatially variable means for change detection and analysis based on inherent SA trends within acquired datasets.

A Moran's I_i scatterplot (Fig. 1) represents SA between an attribute value (x-axis) and the standardized average of neighboring values (y-axis) (Anselin, 1995). Local Moran's I_i can be applied to detect clusters of attribute values (e.g., elevation change) that are high or low relative to the mean change (Ord and Getis, 2001). For geomorphic change detection, hot spots represent clusters of appreciable deposition (Dh) and cold spots indicate clusters of notable erosion (Ec) whereas the remaining quadrants represent outliers of deposition and erosion (Do and Eo, respectively). This method of SA analysis has been used to explore spatial clustering in the formation of desert pavement (Adelsberger and Smith, 2009), linear dune geomorphology (Bullard et al., 1995), variability of coastal dune soils (Kim and Zheng, 2011), patterned ground in permafrost regions (Luoto and Hjort, 2006), and mountain stream morphology (Thompson et al., 2006), but has not been applied to study the dynamics of coastal dunes.

Spatial statistics offer several key advantages over other methods of change detection. First, the results and interpretations are statistically grounded rather than largely descriptive or based on arbitrary thresholds. Second, the analyses are spatially guided or based on the

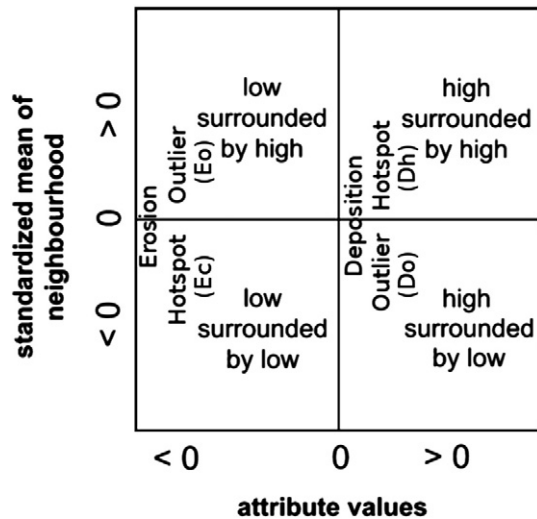


Fig. 1. Moran's I_i scatterplot representing relationships between the attribute value (x-axis) and the standardized average of the neighboring values (y-axis). Modified from Nelson and Boots (2008) to show geomorphic interpretations of significant erosional and depositional hotspot clusters (Ec and Dh, respectively) and outlier values (Eo, Do).

variation of an attribute within a spatial neighborhood in a broader landscape (i.e., they are not limited by a spatially-uniform characterization). Third, related SA measures (e.g., Moran's I_i) can distinguish statistically significant clustering of observations by testing a null hypothesis that observed spatial patterns could have arisen by chance. Thus, rejection of the null hypothesis can be used to define the threshold for spatially correlated change detection (Nelson and Boots, 2008). Use of SA methods in coastal geomorphology is generally lacking, despite the potential for spatial-temporal pattern recognition to provide new insights into underlying process-response relations.

3. Assessing significant geomorphic and sediment budget responses within a mechanically restored coastal dune ecosystem

3.1. Research context and study site

Sandy beach-dune systems are rare in western Canada and comprise less than 10% of the shoreline of British Columbia (Page et al., 2011). Although rare, dune ecosystems are among the most threatened in Canada and provide critical habitat for a plethora of endemic, migratory, and endangered species. Recent interest exists in restoring these ecosystems to improve habitat for species of concern in accordance with provincial and federal legislation (e.g., the Canadian Species at Risk Act, SARA). For instance, in 2009 Parks Canada Agency (PCA) initiated a dynamic dune restoration project at the Wickaninnish Dunes complex in Pacific Rim National Park Reserve on the west coast of Vancouver Island, British Columbia (Fig. 2). The goal of the project was to mechanically remove invasive *Ammophila* spp. (marram grass) to restore more dynamic dune habitat for several endangered species, most notably Grey beach pea vine (*Lathyrus littoralis*, provincially listed) and Pink sandverben (*Abronia umbellata* var *breviflora*, red-listed under SARA). *Ammophila* is well known for its aggressive expansion on foredunes, reduction of biodiversity, and ability to significantly alter foredune sediment budgets and morphodynamics (e.g., Wiedemann and Pickhart, 1996; Wiedemann, 1998; Hesp, 2002). As elsewhere in western North America, *Ammophila* colonized foredunes tend to be steeper and taller, which limits landward sand transport (e.g., Cooper, 1958; Wiedemann and Pickhart, 1996; Wiedemann, 1998). This contrasts foredunes vegetated with native

Leymus mollis (dune grass), which are typically lower, more hummocky, and offer more landward pathways for sand transport (Fig. 3).

The premise guiding the restoration approach of PCA was that removal of invasive marram on the foredune would promote enhanced aeolian activity, increased landward sand transport, and thereby improve habitat for the endangered species. In September 2009, PCA removed vegetation from the foredune using a backhoe equipped with a specialized finger bucket to reduce sand loss during removal. The study site consisted of a 10,320 m² foredune-transgressive dune system (Fig. 4) with 75 m of fully denuded, low (2–3 m) foredune backed by a transgressive dune complex that had a perimeter of 924 m encompassed on all landward boundaries by forest. As such, this site provided a spatially discrete entity for quantifying volumetric and morphological responses to the restoration treatment. Results explored here are for the first year following marram removal.

To establish a baseline control DEM, a bare Earth base map of the study site was derived from airborne LiDAR flown prior to vegetation removal (August 2009). The dataset consisted of over 11,000 point measurements with an average density of 1.13 points m⁻². Subsequent surveys were conducted using a laser total station, approximately bi-monthly between August 2009 and August 2010 (Table 1), and were compared to the LiDAR reference dataset via interpolated DEMs with similar resolution (explained below). A systematic nested data collection strategy (e.g., Chappell et al., 2003b) was used to capture detail of significant features and slope inflection points with grid densities of 0.04 to 0.09 points m⁻² and vertical accuracies (closing errors) of 0.04 to 1.65 cm (Table 1).

The study site was delineated into three discrete geomorphic units: beach, foredune, and transgressive dune complex (Fig. 4) from interpretation of digital orthophotographs collected during the LiDAR mission and ground truthing. The beach unit was approximately 80 m wide and was defined by the seaward limit of the surveys. The landward margin was delimited by the toe of the established foredune and included an ephemeral incipient dune. The foredune unit extended from the beach unit either to the landward extent of foredune vegetation or to the edge of depositional lobes extending from the foredune crest in the baseline LiDAR survey. The transgressive dune unit extended landward from the foredune and was defined by the extent of active sand surface in the initial survey. The area and perimeter of the study site remained essentially constant over the study period. This delineation allowed analysis of sediment budget dynamics (i.e., volumetric changes within and between fixed landform units) and related geomorphic responses to the restoration treatment. Also, as different formative processes control morphodynamics in these units (i.e., swash zone dynamics, aeolian transport, etc.), delineation allowed for refinement of the interpolation method and provided some geomorphic rationale for representative geostatistical modeling. For instance, kriging considers directionality and variation distance (or orientation and size of features, respectively) and delineation allowed for modeling to be compartmentalized into regions where processes and features are mostly related.

3.2. Data and methods

3.2.1. Geostatistical analysis and DEM generation

The spatial structure within the acquired datasets was examined using the geostatistical package 'geoR' (Ribeiro and Diggle, 2001) to produce observed variograms. The data were found to be anisotropic (i.e., distinct directionality was evident), which is consistent with other studies in coastal and aeolian settings (e.g., Swales, 2002; Chappell et al., 2003b). Unfortunately, as the orientation of the geomorphic units was elongated alongshore, insufficient data (i.e., $n = 30$ to 50) were available in the direction of anisotropy (obliquely onshore) to produce directional model variograms. Thus, non-directional model variograms were produced, which is recommended for studies with insufficient data (Chappell et al., 2003a). Gaussian or spherical models

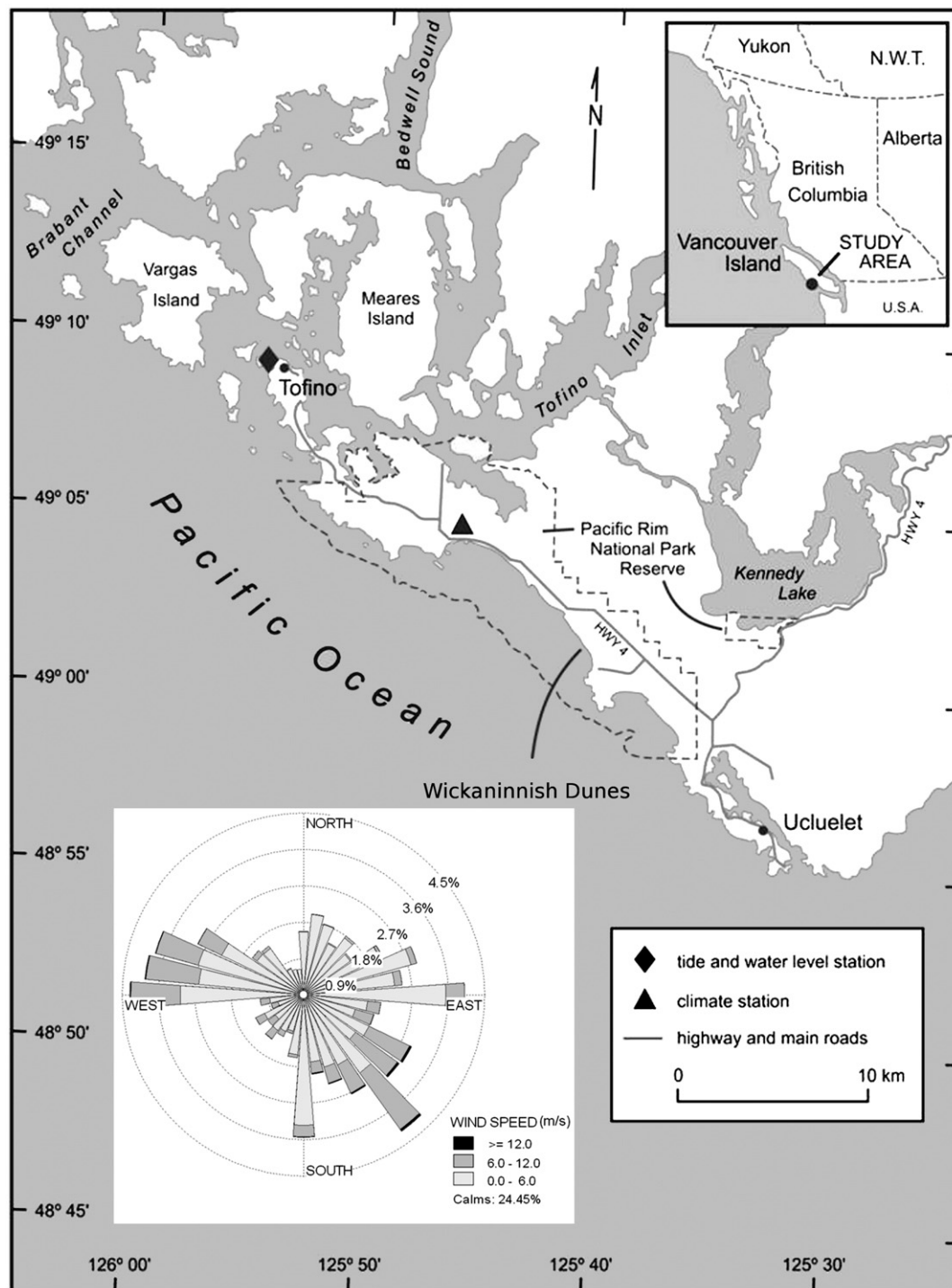


Fig. 2. Map of the study region showing the location of the Wickaninnish Dunes complex within Pacific Rim National Park Reserve near Ucluelet and Tofino, British Columbia, Canada.

best represented the semi-variance in each case (temporally across all three geomorphic units). Model parameters were refined using ordinary least squares to best fit the model to the observed variograms. Combined error in each dataset was determined by the sum of the 95% confidence interval surrounding the mean cross-validation error (based on mean and standard deviation of the cross-validation results) added to the stated instrument precision (0.005 m). Cross-validation analysis showed that the most accurate interpolation occurred in the transgressive dune unit and the most variance occurred in the beach

unit, most likely because of high and low point densities, respectively. The most temporally consistent interpolation (based on mean cross-validation and combined error) occurred in the foredune unit. DEMs of 1-m resolution were then interpolated in geoR using ordinary kriging with weights derived from the model variograms. This resolution was chosen to approximate the same density as the baseline LiDAR data (Table 1) and is within the 1- to 2-meter range suggested as sufficient to capture geomorphic changes within most coastal dune systems (Woolard and Colby, 2002). Cross-shore profiles were also extracted

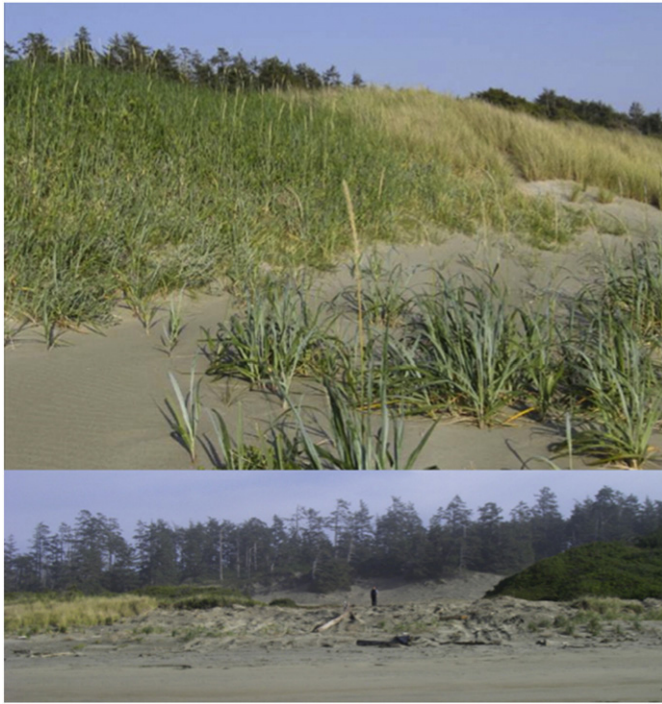


Fig. 3. Upper photograph (taken in July 2009 prior to restoration) shows foredunes in the study area vegetated with invasive marram (*Ammophila arenaria*, upper right) and native dune grass (*Leymus mollis*, upper left). The lower photograph (taken in September 2009 immediately following restoration) shows where vegetation was removed from the foredune. Note person for scale in the middle of removal swath. The site is bordered by a lower, hummockier foredune vegetated with native dune grass backed by forest cover (lower right) and taller, more uniform foredune vegetated with invasive marram (lower, left).

from the DEMs along the transect, shown in Fig. 4, to depict traditional two-dimensional variations in beach-dune morphology over the study period.

Table 1

Summary of data collected to assess landscape responses to mechanical vegetation removal at the study site. Bare sand surface area (10,320 m²) and vegetated perimeter (924 m) of the study site remained essentially constant during the period of study.

Date of survey	Number of points	Point density (pt m ⁻²)	Horizontal closing error (cm)	Vertical closing error (cm)
27 Aug. 2009 (baseline LiDAR)	11,601	1.13	N/A	15.0 (assumed)
24 Sept. 2009	926	0.09	15.2	0.20
23 Oct. 2009	455	0.04	8.0	0.51
8 Dec. 2009	480	0.05	7.8	1.11
15 Jan. 2010	501	0.05	8.2	0.25
5 Mar. 2010	579	0.06	13.4	0.60
13 Apr. 2010	532	0.05	2.2	0.04
30 May 2010	619	0.06	5.7	0.47
8 Jul. 2010	781	0.08	15.0	1.65
14 Aug. 2010	483	0.05	12.7	1.56

3.2.2. Spatial autocorrelation analysis using Moran's I_i

Moran's I_i requires definition of a spatial neighborhood (or weight) that defines the scale over which spatial relationships are calculated (Anselin et al., 2006; Nelson and Boots, 2008). Gridded DEM datasets were imported into the GeoDa™ software (Anselin et al., 2006) to derive spatial weights using assigned threshold distances (TD). This approach (compared to a nearest neighbors method) is appropriate when data are regularly spaced and a theoretical or empirical rationale exists for defining a specific spatial scale (Nelson and Boots, 2008). A spatial weight of 5 m was selected to reflect a spatial scale comparable to dune form and spacing observed within the study site (e.g., depositional lobe and erosional blowout lengths, transgressive dune ridge spacing, etc.). This spatial weight was then used to conduct local Moran's I_i analyses on each change map for the entire study site, using a maximum p-value of 0.05. Statistical significance for Moran's I_i is based on a random permutation procedure that recalculates the statistic many times to generate a reference distribution (Anselin et al., 2006). In this case, a default of 499 permutations was used. Values were then compared to the reference distribution to test the

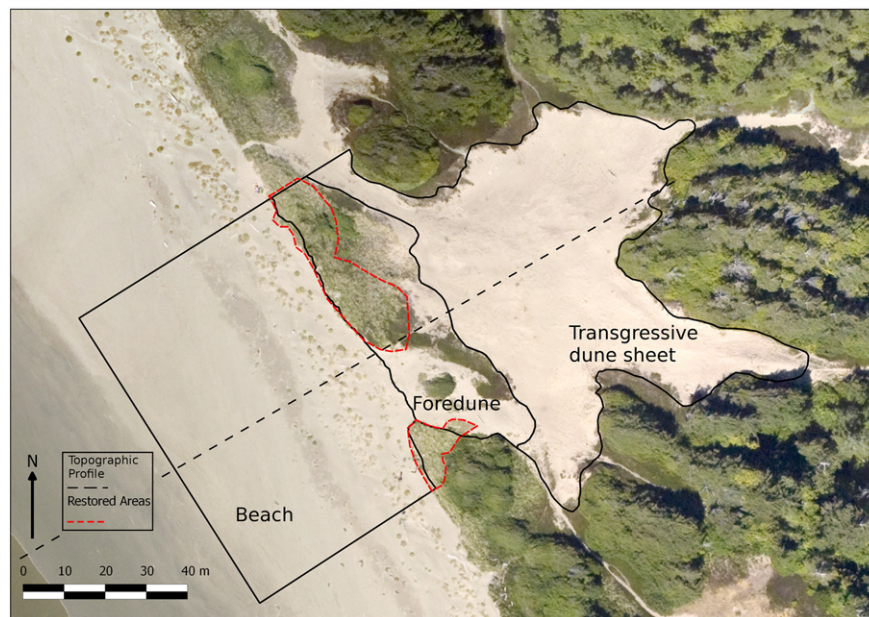


Fig. 4. Aerial photograph of the study site taken during the LiDAR baseline survey (27 August 2009) prior to restoration showing distinct geomorphic units identified for analysis of sediment budget and geomorphic change detection and location of the extracted cross-shore profiles. Areas where invasive marram was removed in the following September are shown by the dashed polygons on the foredune.

hypothesis that surface change was not from chance. Return values, or clusters of extreme deposition and erosion values (Dh and Ec) and outliers (Do and Eo) (Fig. 1) were exported to a GIS to generate maps of local Moran's I_i clusters and remove pixels from surface change DEMs that were not statistically significant. Volumetric changes for each surface change map were then calculated for each geomorphic unit by subtracting the remaining pixels between successive time intervals. Volumes were then normalized (divided) by the total area of each unit to yield area-normalized volumetric changes (in $\text{m}^3 \text{m}^{-2}$) to provide an indication of effective depth of sediment accretion (+) or erosion (–).

3.3. Results

3.3.1. Geomorphic changes from extracted cross-shore profiles

Extracted cross-shore profiles (Fig. 5) show variations in two-dimensional beach-dune morphology during the period of study. Significant accretion occurred within the beach unit in the fall months from bar welding and berm development, followed by beach erosion and steepening during the winter when the incipient dune was eroded and the lower seaward slope of the foredune was scarped and retreated by approximately 2 m. A distinct pivot point, or node of convergence in slope profiles where little change occurs, was evident near the erosional threshold elevation at 5.5 m above chart datum (maCD) (Beaugrand, 2010). This marks a notable change from a depositional setting on the beach to an erosional one on the foredune and results from combined wind erosion and scarping by high wave runoff between October and March. Accretion resumed in March through April resulting in as much as 0.5 m of accretion on the upper beach, incipient dune and foredune toe region by May.

Appreciable aeolian activity occurred near the crest of the foredune throughout the year as demonstrated by fluctuations in erosional and depositional surface changes. Erosion of approximately 0.25 m occurs here in October, most likely in response to sediment losses during and immediately following vegetation removal. The crest then built quickly through winter and spring by approximately 0.5 m ($+0.23 \text{ cm day}^{-1}$) to the May survey, then eroded over the summer by about 0.2 m ($-0.26 \text{ cm day}^{-1}$) by August to a height very similar to that prior to restoration (Fig. 5). Most accretion within the foredune occurred within depositional lobes that extended landward from the crest. Extension of these lobes, combined with some (1 to 2 m) progradation on the lower stoss and dune toe region,

resulted in a wider foredune profile compared to the pre-restoration LiDAR survey. This response corresponded with enhanced supply to the upper beach and incipient dune from April through August. The mid-stoss slope (between 7 and 7.5 m aCD) was a location of sediment bypassing and minimal morphological change throughout the year while sediment from the beach and lower stoss slope moved landward to rebuild and steepen the crest and feed expanding depositional lobes. By August, nearly one year following the restoration treatment, the seaward portion of the foredune profile attained a very similar height and form to that of the pre-restoration profile whilst remaining essentially free of vegetation.

The transgressive dune complex experienced slight vertical accretion immediately following restoration treatment in September with peak accretion evident in May (Fig. 5). This occurred in response to improved sand transport pathways over the foredune following rebuilding of the scarped seaward slope, high inputs from the depositional lobes extending from the foredune, and by migration of small, discrete dunes that eventually fed into the precipitation ridge on the eastern edge of the complex, as also shown in the change detection surfaces described below.

3.3.2. Significant volumetric and geomorphic changes derived from Moran's I_i

Areal coverage of significant clusters of deposition (Dh) and erosion (Ec) and associated sediment volume changes throughout the year for each geomorphic unit are summarized in Table 2. Spatial-temporal maps of related Moran's I_i change cluster surfaces are depicted in Fig. 6.

Immediately following the restoration treatment in September 2009, 64% (6574 m^2) of the surface area of the site experienced significant erosional (33%) or depositional (31%) change (Table 2). In terms of normalized sediment volumes, most of the deposition occurred in the beach unit ($+0.167 \text{ m}^3 \text{m}^{-2}$) while the foredune experienced net erosion ($-0.020 \text{ m}^3 \text{m}^{-2}$). By October, erosional area increased to 37% while depositional area dropped to 24%. The significant change cluster map (Fig. 6) for October shows pronounced deposition on the lower beach and in the incipient dune zone fronting the foredune with some localized erosion of the incipient dune. Corresponding volumetric responses indicate that most of the accretion also occurred within the beach unit during this time ($+0.214 \text{ m}^3 \text{m}^{-2}$) by almost an order of magnitude greater than in the other geomorphic units (Table 2).

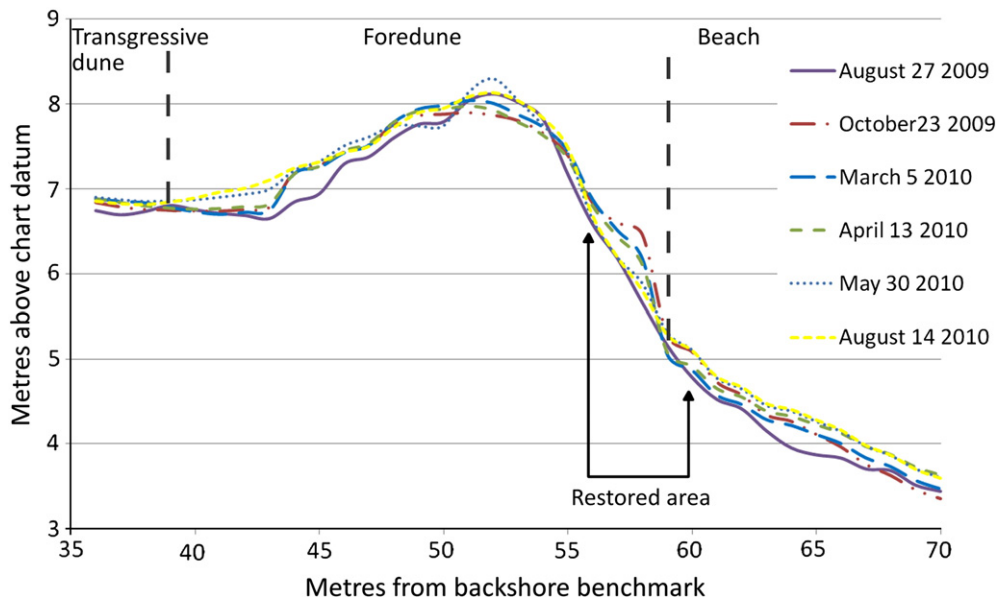


Fig. 5. Cross-shore topographic profiles extracted from interpolated DEM data for select survey dates.

Table 2

Results of local Moran's I_i change detection for a spatial weight of 5 m. Areal coverage of statistically significant depositional and erosional clusters (Dh and Ec, respectively) and outlier values (Do, Eo) are shown on the left side of the table. Estimates of statistically significant sediment volume changes and area-normalized values show average sediment accretion (+) or erosion (−) within each unit, as shown on the right side of the table. All values are referenced against the initial LiDAR baseline survey on 27 August 2009.

Date (days after baseline survey)	Area of coverage in m ² (% total site area)				Volumetric change from LiDAR baseline in m ³ (m ³ m ^{−2})		
	Dh	Do	Eo	Ec	Beach 4470 m ²	Foredune 1565 m ²	Transgressive dune 4285 m ²
24 Sept. 2009 (31)	3185 (31)	319 (3)	187 (2)	3389 (33)	+746 (+0.167)	−31 (−0.020)	+187 (+0.044)
23 Oct. 2009 (57)	2505 (24)	450 (4)	130 (1)	3859 (37)	+956 (+0.214)	+45 (+0.029)	+155 (+0.036)
8 Dec. 2009 (103)	2552 (25)	422 (4)	365 (4)	3355 (33)	+483 (+0.108)	+127 (+0.081)	+221 (+0.052)
15 Jan. 2010 (141)	3967 (38)	107 (1)	517 (5)	2181 (21)	−133 (−0.030)	+124 (+0.079)	+302 (+0.071)
5 Mar. 2010 (190)	5135 (50)	70 (1)	491 (5)	2752 (27)	−650 (−0.145)	+159 (+0.102)	+368 (+0.086)
13 Apr. 2010 (229)	3406 (33)	366 (4)	152 (1)	3567 (35)	+766 (+0.171)	+31 (+0.020)	+270 (+0.063)
30 May 2010 (276)	4260 (41)	171 (2)	226 (2)	3626 (35)	+365 (+0.082)	+116 (+0.074)	+348 (+0.081)
8 Jul. 2010 (315)	3083 (30)	373 (4)	224 (2)	3540 (34)	+870 (+0.195)	+198 (+0.127)	+112 (+0.026)
14 Aug. 2010 (352)	2980 (29)	369 (4)	314 (3)	3551 (34)	+863 (+0.193)	+162 (+0.104)	+287 (+0.067)

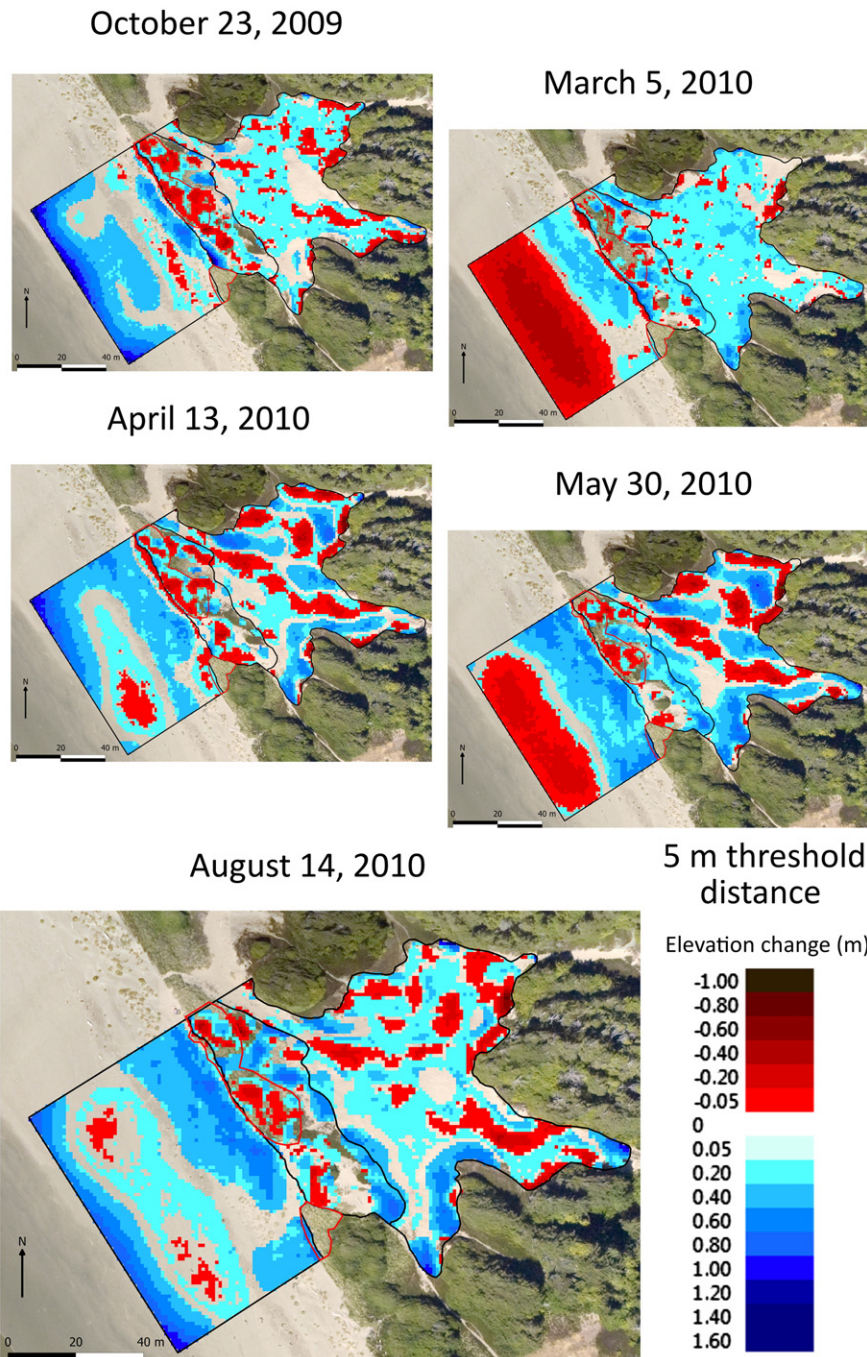


Fig. 6. Significant change surfaces derived from Moran's I_i cluster identification for a 5 m spatial weight. Calculated surface areas and volumes are shown in Table 2.

By December, the study site experienced more significant erosion than deposition (33 vs. 25%) although volumetric changes showed that all geomorphic units experienced net positive sediment inputs. Accretion increased in the foredune and transgressive dune units while it declined on the beach to approximately 50% that in October. From January through March, deposition reached a maximum coverage of 5135 m² (50%) in March nearly doubling erosional coverage (27%). Most of this occurred as volumetric losses in the beach unit during this time (Table 2), particularly on the upper intertidal beach (Fig. 6), while significant deposition occurred on the supratidal beach fronting the foredune. The highest and most widespread deposition within the transgressive dune also occurred during this time.

In April, depositional area declined markedly from the March maximum to a value comparable to erosional area (33 vs. 35%, respectively). Accretion was more widespread, however, on the beach than in March with +0.171 m³ m⁻² of sediment deposited to replenish eroded material in the intertidal zone and with further delivery to the supratidal zone and incipient dune (Fig. 6). Distinct erosional zones appeared, however, on the seaward face and crest of the foredune and within several deflated zones in the transgressive dune. These erosional zones were all bounded by depositional areas. Following net erosion throughout the winter into early spring (January to March), the beach re-established a positive sediment budget and accretion occurred most rapidly between March and April at a rate of approximately +36 m³ day⁻¹. This deposition resulted in as much as 0.5 m of accretion on the upper beach, incipient dune and foredune toe region by May (Figs. 5, 6).

By late spring (May), depositional surface area increased appreciably to 41% while erosional area remained essentially constant (35%). On the beach, extensive accretion was evident on the supratidal beach and incipient dune zone (Fig. 6), although depositional volumes declined overall to 48% of April values (Table 2). The foredune experienced a significant increase in sediment volume (3.74 times that in April) mostly through distinct lobes that grew and extended from the crest into the transgressive dune complex. The transgressive dune unit also saw notable increases in accretion as discrete dunes within the complex and by continued growth of a precipitation ridge on the eastern margin of the complex (Fig. 6). By July, depositional surface area fell below erosional area (30 vs. 34%) although the beach unit saw the most significant amount of volumetric increase (+0.195 m³ m⁻²) since October. Coincidentally, the foredune saw its highest increase in sediment volume (+0.127 m³ m⁻²) while the transgressive dune experienced a significant drop in accretion to its lowest value (+0.026 m³ m⁻²). Proportional depositional and erosional surface areas remain essentially constant through late summer

(August) with percent coverage values similar to those following restoration treatment nearly a year earlier showing slightly more erosional area over depositional (34 vs. 29% Dh). Depositional volumes declined only slightly on the beach, whereas the foredune experienced a more notable decline and the transgressive dune unit saw an appreciable increase of approximately 2.6 times the depositional volume in July (Table 2).

4. Discussion

4.1. Sediment budget and geomorphic responses following restoration treatment

Fig. 7 shows the trends in area-normalized sediment volume (Table 2) and provides insight into sediment budget responses within and between landform units for the year following the restoration treatment. Compared to the pre-restoration survey, all geomorphic units experienced significant net positive gains in sediment volumes during the year following the restoration treatment. The beach received the highest normalized amount (+0.193 m³ m⁻² or 2.45 m³ day⁻¹), indicating an appreciable supply of sand delivered to the system, while the foredune experienced a higher normalized volume but lower rate of sand input (+0.104 m³ m⁻² or 0.460 m³ day⁻¹) compared to the transgressive dune (+0.067 m³ m⁻² or 0.815 m³ day⁻¹). The beach unit was the most dynamic as shown by the highest amount of erosion, greatest total (differential) volumetric change (see March and April values in Table 2), and highest net annual sediment budget. The appreciable supply of sand to the beach was available for landward delivery to the dune systems and was essentially independent of the restoration effort.

The positive annual sediment budget of the smaller foredune unit was approximately 19% that of the adjoining beach. The foredune experienced erosion immediately following vegetation removal in September followed by slight accretion in October, which increased to a maximum in March. Although significant erosion occurred on the lower stoss slope and dune toe during the fall, partly in response to disturbance during vegetation removal and partly from wave scarping in the early winter, significant accretion occurred over 1.87 times more surface area of the foredune than erosion by March. This was manifested by deposition on the lower stoss slope (scarp rebuilding) and by depositional lobes extending from (and rebuilding) the crest into the immediate lee (Fig. 6). Following a major erosive event on the beach in March and aeolian deflation on the foredune, the sediment budget of the foredune unit declined appreciably by April compared to the pre-restoration survey. From April through

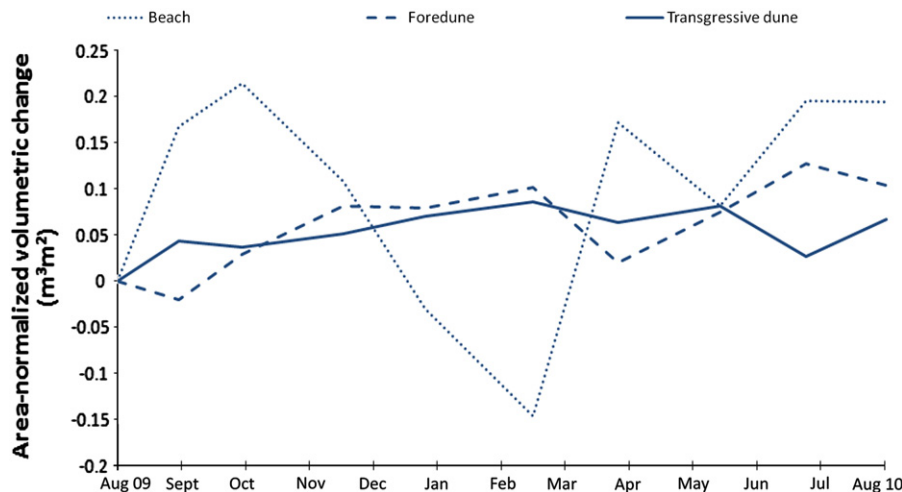


Fig. 7. Comparison of sediment budget responses across geomorphic units during the year following restoration treatment derived using Moran's *I*. Area-normalized volumetric changes (m³ m⁻²) provide an indication of effective depth of sediment accretion (+) or erosion (-).

August, the sediment budget of the foredune increased progressively despite a notable drop in sediment inputs to the beach in late spring to early summer (April through May, Fig. 7). Erosion is widespread over the surfaces of the foredune and transgressive dune units throughout this time (Fig. 6) despite net positive increases in sediment budgets.

The transgressive dune complex is the only unit within which geomorphic and sediment volume responses are controlled almost entirely by aeolian activity and it responds essentially as a sink for sands delivered through and/or from the restored foredune. Immediately following restoration in September, accretion in the transgressive dune was positive and significantly greater than the foredune, which displayed net erosion during this time (Fig. 7, Table 2). A high influx of sediment to the beach in October, coincident with deflation of the foredune crest and extension of depositional lobes into the lee (Figs. 5–7), suggests that sand bypassing through the foredune was most pronounced in the two months immediately following restoration treatment. After this, the sediment budget of the transgressive dune complex increases to a maximum in March followed by a slight decline to April then a rise into May. This trend is coincident with (although less pronounced than) that of the foredune unit, which may reflect a recoupling of accretion responses between the foredune and transgressive dune units during a time when inputs from the beach increase rapidly. Maps of change detection (Fig. 6) and cross-shore profiles (Fig. 5) show that peak accretion in the transgressive dune complex occurs in response to inputs from depositional lobes extending from the foredune and from smaller dunes that develop and migrate within the complex toward a growing precipitation ridge along the eastern edge of the complex. These responses are concurrent with improved pathways for sand transport over the foredune following rebuilding of the scarped seaward slope by the natural processes of aeolian accretion.

Thus, in the year following restoration treatment, significant amounts of sediment were delivered to the foredune, which recovered rapidly to mechanical disturbance during vegetation removal and subsequent natural erosion events, and reformed a seaward slope profile very similar to that of the pre-restoration surface. During this morphodynamic recovery, the foredune also experienced aeolian deflation, sediment bypassing, and landward extension, which resulted in a net positive sediment budget of the transgressive dune complex.

4.2. Methodological and logistical considerations

As discussed in Section 2, the accuracy of DEMs used for the analysis of change detection is constrained by spatial and temporal resolution as well as sampling strategy. In terms of data pre-processing, selection of an appropriate grid resolution is key for avoiding the modifiable area unit problem (MAUP, Openshaw, 1984) where arbitrarily defined gridded results can suffer systematic bias in, or masking of, inherent spatial patterns. In this case, the 1 m-grid spacing was chosen to approximate the density of the baseline LiDAR survey (Table 1) and is suggested to be sufficient to capture morphological and volumetric changes within coastal dune systems (Woolard and Colby, 2002). For ground-based surveys, a systematic nested data collection strategy is recommended (cf. Chappell et al., 2003b) that captures sufficient topographic variations and features (e.g., landform margins, slope inflection points).

In terms of analytical methods, the benefit of applying a spatial autocorrelation method, such as local Moran's I_i , over more spatially uniform threshold-defined approaches (e.g., using instrument precision or percentage change thresholds) is the ability to detect change processes within a neighborhood of related observations and interpret resulting spatial patterns. As the size of a spatial neighborhood can be defined (using spatial weights) using theoretical rationale or empirical evidence on underlying spatial structure in the dataset

(e.g., variogram results or observed landform length scales), spatial autocorrelation methods provide a means to explore varying spatial scales and test hypotheses of what controls observed changes. For example, a spatial weight of 5 m was used in this study, based on average dune feature size and spacing at the site. A smaller spatial weight of 1.5 m, chosen to scale with the aeolian saltation process, was explored by Eamer and Walker (2013) at the same study site, but this scale of analysis tended to buffer extreme results (i.e., less extensive Ec and Dh) and produced consistently lower volumetric changes and more sporadic, localized patterns of change. As the temporal scale of saltation (minutes to hours) does not correspond with the time intervals of the topographic surveys, it should not be expected that net sand movements over months would be better characterized by this smaller spatial scale. The 1.5 m spatial weight did, however, better represent change associated with smaller, hummocky features in the incipient dune zone and erosive blowouts on the foredune (Eamer and Walker, 2013). Generally, however, the larger 5 m spatial weight generated more contiguous patterns of change that matched qualitative observations at the site. Technically, a range of other spatial scales could be explored either by examining observed variograms or based on modal dimensions of various processes or landforms of interest. The finest scale would be constrained to some multiple of the resolution of the dataset. The largest scale (without becoming global or spatially uniform) would be slightly smaller than the extent of the study site; keeping in mind that edge effects (resulting from missing values along the search perimeter) increase for larger spatial neighborhoods.

The spatial structure of most geomorphic datasets is also typically anisotropic (Oliver et al., 1989b; Swales, 2002; Chappell et al., 2003a, 2003b), meaning a distinct directionality occurs in the variation of surface properties. For instance, the degree of morphological variation in a relatively uniform beach backshore is typically less alongshore compared to that in the onshore direction where a greater variation in elevation and other properties arise toward the foredune. Anisotropic variation can also result from different processes (e.g., wave swash vs. aeolian transport) operating at differing spatial and temporal scales and in varying directions, which highlights the benefit of subdividing broader landscapes into distinct geomorphic units. A general sampling guideline of 30 to 50 data points in the hypothesized or observed direction of anisotropy is recommended for generation of sufficient directional model variograms, otherwise, non-directional variograms should be used (Chappell et al., 2003a).

Finally, to most accurately assess the impacts of restoration treatments (e.g., complete or partial removal of vegetation) on dune sediment budgets and morphodynamic responses, contemporaneous control data are useful. Ideally, this would involve concurrent data collection within an adjacent site of identical geomorphology where-in the treatment was not applied. In this case study, an identical site did not exist as the implemented treatment was predetermined by PCA and a vegetated (control) swath of the foredune was not left for experimental purposes. Instead, interval survey datasets were normalized against the baseline LiDAR DEM to produce volumetric changes that were relative to pre-restoration conditions. Contemporaneous spatial control should also, ideally, be complemented by temporal control data including longer-term (pre- and post-restoration) observations of: i) volumetric, areal and geomorphic changes from land surveys and historical aerial photographs, and ii) process indicators, such as the occurrence of transporting winds and saltation activity. These data provide ambient conditions and temporal trends or anomalies in process–response dynamics in the system. Pragmatically, incorporating such extensive control considerations is most often logistically constrained (e.g., lack of longer-term datasets, no comparable co-located sites, etc.) and requires careful planning and implementation of treatment and data collection protocols with multi-year monitoring commitments.

4.3. Assessing effectiveness of dynamic restoration treatments

For successful restoration of a coastal dune system from a stabilized state dominated by invasive vegetation, one key ecological aim is to promote a more resilient native dune plant community. Given the fundamental role of natural disturbance in dune ecosystems, this requires morphodynamic changes that re-establish nutrient-limited soil conditions and processes of wind erosion, abrasion, and sand burial, that favor early successional (pioneer) species over later successional and non-native species, which often form monocultures (van Zoest, 1992; Arens and Geelen, 2006; Nordstrom, 2008). The result is essentially a more resilient dune ecosystem that holds greater ecological value than that of a stabilized system (Nordstrom, 2008). To this end, several key geomorphic indicators are identified here that could be used to assess the effectiveness of implemented treatments in enhancing favorable morphodynamic conditions for dune ecosystem restoration. These include: i) increased aeolian activity (sand transport, erosion, deposition), ii) enlarged active sand surface areas, iii) a positive sediment budget, iv) increased dune morphodynamics, v) improved geomorphic diversity, and vi) enhanced geomorphic resilience.

First, increased aeolian activity in dune systems limits pedogenic development and, when enhanced wind erosion removes developed topsoils, helps re-establish nutrient-limited soil conditions on previously stabilized soil surfaces. In turn, processes of aeolian disturbance (e.g., wind erosion, abrasion, and sand burial) enhance growth conditions for early successional dune species over later successional and non-native species (van Zoest, 1992; Arens and Geelen, 2006; Nordstrom, 2008). Related, the second indicator of enlarged active sand surface area provides evidence that aeolian activity is increasing and that stable topsoil surfaces have either been eroded or buried. Prior to the restoration treatment (between 1973 and 2007), Heathfield and Walker (2011) documented a 28% decline in active sand surface area within the larger dune complex at Wickaninnish and suggested that colonization by *Ammophila* spp. on the foredune and encroachment by native Kinnikinnik (*Arctostaphylos uva-ursi*) and Sitka spruce (*Picea sitchensis*) on the transgressive dune perimeter were responsible. In the year following restoration, active sand surface area increased only slightly (i.e., by 570 m² or approximately 5.5% of the total site area) mostly by creation of open sand surfaces where vegetation was removed on the foredune. Evidence also exists of expansion of the precipitation ridge along the eastern edge of the study site, but this was mostly a volumetric increase that entailed very little expansion in sand surface area.

On the third indicator (positive sediment budget), dune systems colonized by invasive *Ammophila* often experience alterations in sediment budgets that result in reduced landward sand transfers and foredune steepening (e.g., Wiedemann and Pickhart, 1996; Wiedemann, 1998; Hesp, 2002). Although foredunes in such instances may continue to experience a positive sediment budget, as they grow and/or steepen in response, consequent declines may occur in the sediment budget of blowout, transgressive, and/or parabolic dunes landward of the foredune. Following restoration treatment, all geomorphic units within the beach-dune complex at Wickaninnish Beach showed significant increases in sediment volumes in response to the combined effects of high sand supply to the beach (source) and reduced vegetation cover on the foredune (source and sink), which improved sand delivery to the transgressive dune complex (sink). This response is clearly related to removal of vegetation from the foredune and enhanced pathways for sand transport into the transgressive dune as shown by expanding erosional blowouts and depositional lobes in the lee of the foredune crest. For this response to continue and promote increased active sand surface area in the transgressive dune unit, however, a maintenance program for removal of re-establishing *Ammophila* will be required.

The fourth indicator of increased dune morphodynamics relates to the enhanced development of existing dune forms within a

restoration site. Examples include increased foredune growth; dune progradation and/or landward migration; and greater development and extension/migration of blowouts, sub-parabolics, smaller transgressive dune forms (e.g., transverse or barchanoid ridges) and/or precipitation ridges along vegetated margins. In this case study, all of the aforementioned examples occurred within the first year following vegetation removal as a result of enhanced landward sand transfers and positive sediment budgets. Related, the fifth indicator of improved geomorphic diversity is manifested as an increase in the number and/or type of aeolian features at a site. Examples from this case study included, new and extending erosional blowouts and depositional lobes in the foredune crest region and new migrating dunes and deflation basins within the transgressive dune complex (Figs. 5–7).

The final, and perhaps most important, indicator of morphodynamic effectiveness is enhanced geomorphic resilience as shown by recovery of dune morphology to some pre-disturbance state. Inherent to the geomorphology of coastal dune systems are periodic erosion–stabilization cycles (Arens and Geelen, 2006; Nordstrom, 2008). For example, foredunes on active coastlines with a stable to positive sediment budgets will often rebuild quickly following coastal erosion events (e.g., Psuty, 1988; Hesp, 2002; Ollerhead et al., 2013), as is evident at the study site (Beaugrand, 2010; Heathfield and Walker, 2011). Geomorphic resilience and related erosion–recovery cycles were demonstrated to the pre-restoration state as shown by recovery of the seaward portion of the foredune profile to a form very similar to that pre-restoration (Fig. 5). What is more challenging to assess, however, is whether or not foredunes colonized with invasive species can recover to a more ‘natural’ eco-geomorphic state. In other words, can dunes restored by the removal of invasive vegetation recover to some morphodynamic state that existed prior to colonization by the invasive species? Given a lack of comparable longer-term morphological observations, which is not uncommon in western Canada and the United States, this question remains unanswered by the current case study. The morphology of small remnant stretches of native dune ecosystems in the study region and elsewhere in British Columbia, however, suggests that dunes vegetated with native *Leymus mollis*, which has a less dense growth pattern, tend to be less steep, more hummocky, and more dynamic in geomorphic response (Page et al., 2011) (Fig. 3).

The results of the case study, although temporally limited to a single year, suggest that the implemented treatment of restoration was effective in promoting several positive indicators of morphodynamic recovery including: increased aeolian activity, positive sediment budgets (Table 2, Fig. 7), increased dune morphodynamics, enhanced geomorphic diversity and demonstrated geomorphic resilience. Should these responses continue, it is anticipated that dune mobility and active sand surface area at the study site would increase. Given subsequent observations of rapid re-establishment of *Ammophila* at the site, however, continued maintenance pulling will be required to preserve this enhanced morphodynamic state. This is confounded by a broader conundrum as to what the longer-term eco-morphodynamic state of restoration is (e.g., pre-invasive species colonization). For Parks Canada, it is simply a dynamic landscape that will support re-establishment of viable populations of endangered species so as to satisfy the Species at Risk Act obligations. Observed rates of historical dune stabilization at the site (Heathfield and Walker, 2011) and regional climate trends toward wetter, warmer conditions in coastal British Columbia (Walker and Sydneysmith, 2008) suggest that the longer-term eco-geomorphic state is most likely a fully stabilized (forested) ecosystem. As such, the current restoration state may not be on trajectory with the longer-term eco-geomorphic state of dune systems in the region. Thus, without frequent restoration intervention, complete stabilization of the transgressive dune systems could occur within a few decades (Heathfield and Walker, 2011).

5. Conclusions

A shift from dune restoration for stabilized landscapes toward more dynamically functional ecosystems is underway. Dynamic dune restoration projects, such as the case study reported here, aim to re-establish disturbance processes, such as wind erosion, abrasion, and sand burial, to promote more resilient landscapes with required ecological conditions and gradients needed to support natural ecological communities. Such projects provide opportunities to explore landscape responses to implemented treatments and, in turn, assess effectiveness. This paper reviews established spatial statistical methods used in other areas of geomorphology and natural sciences to quantify and examine significant spatial–temporal change patterns. One particular method, local Moran's I_i , is applied to detect and assess the impacts of the removal of invasive vegetation from a foredune-transgressive dune complex. The key findings of this study include:

- 1) Over the first year following restoration treatment, all three geomorphic units experienced net increases in sediment budgets as compared to the pre-restoration surface. Despite seasonally dynamic response and independent of the restoration effort, the beach received a positive net annual supply of sand that was available for landward transport to the dune systems. Following slight erosion during vegetation removal, the foredune showed consistently positive sediment volumes and rapid recovery to the initial disturbance as well as subsequent erosive events. The seaward slope profile recovered rapidly to a form similar to that pre-restoration whereas the foredune also experienced aeolian deflation, blowout development, and landward extension of depositional lobes into the transgressive dune complex. Sand bypassing through the foredune was most pronounced in the two months following restoration treatment and by spring volumetric trends within the foredune and transgressive dune complex appear to synchronize, or recouple, during a time when inputs from the beach increased most rapidly. Peak accretion in the transgressive dune results from depositional lobes extending from the foredune, erosion of deflationary areas that feed smaller dunes migrating within the complex, and growth of a precipitation ridge along the eastern margin of the complex.
- 2) Various methodological and logistical considerations for detecting significant changes in dynamic dune landscapes are recommended. These include care in the collection and pre-processing of topographic data (e.g., sampling strategy development, handling of uncertainty, consideration of inherent spatial relations and structure, data normalization), incorporation of control data as possible, and application of an analytical method that considers local spatial autocorrelation, such as Moran's I_i . Spatial statistics are recommended over more traditional, spatially uniform approaches to provide a more robust means for detecting processes of local change and resulting spatial–temporal patterns that incorporate the influence of spatial autocorrelation and scale and, in turn, allow for hypothesis testing on key process–response dynamics that may be driving observed patterns of change.
- 3) To better assess the effectiveness of dynamic restoration methods, several key geomorphic indicators are recommended, including: i) increased aeolian activity, ii) enlarged active sand surface area, iii) positive sediment budgets, iv) increased dune morphodynamics, v) improved geomorphic diversity, and vi) enhanced geomorphic resilience. These indicators collectively reflect important morphodynamic changes that are required to re-establish ecological conditions and processes that favor a more resilient ecosystem comprised of more native dune species over later successional and/or non-native species.

Although limited in temporal scope, the results of this case study indicate that the implemented restoration treatment was effective in promoting all indicators except for increasing sand surface area.

Given decadal scale observations of climatic changes and longer-term eco-geomorphic trajectory toward stabilization in the region, however, it is unlikely that the geomorphic effectiveness of this restoration effort will continue without frequent treatment interventions.

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